

B. Tech Degree IV Semester Examination April 2011

ME 403 ADVANCED MECHANICS OF SOLIDS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Define plane stress and plain strain. Give examples.
- (b) State and explain St. Venant's principle.
- (c) Represent all the nine stress components on a cubical element with positive directions.
- (d) Show the variation of σ_r and σ_θ for a cylinder subjected to (i) external pressure (ii) internal pressure.
- (e) State and explain the concept of virtual work.
- (f) State Castigliano's theorems.
- (g) Write short notes on (i) shear flow (ii) shear centre.
- (h) Explain membrane analogy.

PART B

(4 x 15 = 60)

- II. (a) Explain the construction of a 2-D stress problem. (8)
 - (b) Write notes on strain measurement. (7)
- OR**
- III. (a) Explain compatibility conditions. (8)
 - (b) Explain stress function with examples. (7)
- IV. (a) Obtain the general expression for σ_r and σ_θ for a rotating disk of uniform thickness. (8)
 - (b) Derive the general equilibrium equations in polar coordinates. (7)

OR

- V. A flat steel disk of 75cm outside diameter with 15 cm diameter hole is shrunk around a solid steel shaft. The shrink fit allowance is 1 part in 1000 (ie, an allowance of 0.0075cm radius). $E = 214 \times 10^6$ kPa.
 - (i) What are the stresses due to shrink fit?
 - (ii) At what rpm will the shrink fit loosen up as a result of rotation? (15)

- VI. Write notes on (i) Stress invariants (ii) Reciprocal theorems in energy methods (iii) Octahedral shear stress. (15)

OR

- VII. Find the principal stresses and check for invariance for the following case.

$$\tau_{ij} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -2 & -3 \\ 1 & -3 & 4 \end{bmatrix} \quad (15)$$

- VIII. Derive an expression for stresses acting on a bar of elliptical cross section. Also find the maximum stress, angle of twist and torsional rigidity. (15)

OR

- IX. Prove that $\theta = \frac{q}{2AG} \oint \frac{ds}{t}$ for thin walled tubes. (15)